

Sanctuary: Mission to the moon

Celebrating Apollo 11's anniversary, a team of European sites will undertake the very first privately-funded 'mission to the moon'. <https://digit.in/apollo11anni>



Drones will decide who to kill

The US Government announced that they are developing drones that identifies targets using AI algorithms. <https://digit.in/drones>

The setup in the TIFR experiment was relatively simple, and involved a glass target in vacuum chamber on a tabletop. Conventionally, such experiments require the use of enormous particle accelerators. A high power laser was focused on a spot of a few micrometre radius on a glass target. The electrons in the glass target were almost immediately kicked out at speeds approaching that of the speed of light in free space. The intensity was enough to create millions of amperes of current pulses. The electron pulses were extremely short, as the light pulses exciting them were also extremely short, in the range of tens of femtoseconds. Since light slowed down within the glass, the electrons moved faster than light within the medium. Because of this, the electrons emitted the Cherenkov radiation. Measuring the radiation, allowed the researchers to find out the number of FTL electrons, as well as for how long they last within the medium.

The Cherenkov radiation lasted for extremely short durations at the TIFR tabletop experiment. So short, that it was beyond the measuring abilities of current electronic circuits. The team used an ultrafast shutter triggered by the same laser to measure the duration of the radiation. The shutter remained open for two picoseconds, and allowed for measuring the radiation with a temporal resolution that was a thousand times better than conventional approaches. The observations showed that the electrons actually lasted longer than theoretical predictions. In fact, they lasted longer than the time estimated for them to cross the medium. The electrons actually lasted for about 2,000 times the duration of the laser pulse. The researchers from CELIA are investigating aspects of the unexpected behaviour.

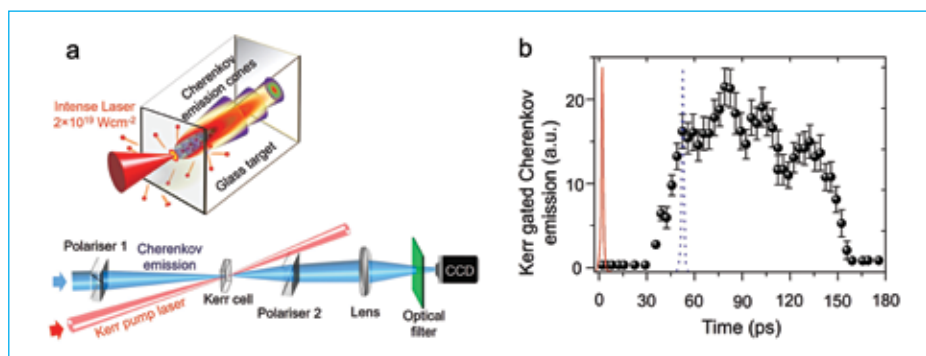
The high speed electrons can be useful in generating some of the most exotic states of matter known. Researchers can get to peek into condi-

tions similar to the hearts of stars and planets, in a lab environment.

To understand more about the experiment and the exotic states of matter created within the glass, we reached out to Professor G Ravindra Kumar from TIFR. At such high intensities, matter at the focal point of the high intensity laser can be subject to tremendous amount of pressure and get heated at the same time. The only place in the Earth where matter is subject to such high temperatures and pressures, is within the core of the planet. In the atmosphere or on the surface where humans live, such extreme conditions do not exist naturally. The matter is crushed to such an extent that most normal forms of

and acquire the properties of metals. Crystals can also be squeezed between two diamond needles to convert them into a metallic state. A laser is a simpler way to study these materials under extreme conditions. Apart from the high temperature and high pressure, there is also an extreme amount of compression, and shock waves within the material. For all these reasons materials are pushed into states that they normally do not experience. More information can be found in the article titled "Exotic States of Matter" in the April 2018 issue of Digit.

The research can advance development of x-ray, ion and electron sources for medical and industrial applications. The research can also potentially



The setup used in the TIFR experiment

crystal will not survive the temperature and pressure. The atoms themselves are squeezed together. "When we squeeze atoms together," Professor Ravindra Kumar explains, "the atomic arrangement in a solid undergoes what are called 'phase transitions', i.e. from one phase to another. Imagine something happening under millions of atmospheres of pressure, or hundreds of millions of atmospheres. It is literally crushing things in such a way, that the matter has to settle into some new forms. The atoms are squeezed together so tightly, that normal structures do not make sense at all at that density and pressure." Normally, glass or plastics are insulators, they do not conduct heat or electricity very well. If you fire one of the lasers with the high temperatures and pressure at glass or plastics, they actually start conducting

help in developing laser driven fusion. There may be many applications in the future, but for the present the experiment was conducted to satisfy basic human curiosity. Professor Ravindra Kumar says, "Technology is recent, but scientific knowledge is much older. It is the scientific knowledge of hundreds of years which has triggered all the major changes in our lives, for the better. The most basic thing that the human mind wants to seek is knowledge about the world that we live in. It is not just our privilege, but it is our duty to understand nature. The internet, technology and the modern life as we know it would not have been possible if we did not understand silicon, or if we did not understand how electrons in solids behave. Science drives technology. The core of our existence is to understand the universe in which we were born." 